Yoonjae Hwang

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EDUCATION

University of Southern California 08/2025 – Present
Master of Science in Computer Science

University of Illinois at Urbana Champaign 08/2017 – 05/2023
Bachelor of Science in Materials Engineering; Minor in Computer Science

RESEARCH INTERESTS

Multi-Agent Simulation, Digital-Twin Technology, Distributed Systems, Efficient Machine Learning

RESEARCH EXPERIENCE

Sim-to-Real Transfer for Robust Robot Manipulation via Synthetic Data Generation 05/2025 – Present
University of Southern California | Personal Research

- **Designing a sim-to-real transfer pipeline** for robotic manipulation using **ManiSkill3** (GPU-parallelized simulator), generating large-scale synthetic demonstration datasets via motion planning and reinforcement learning
- **Constraining robot action space** to 4-DOF (xyz translation + gripper) and trained policies using **PPO/SAC on a V100 GPU** cluster, achieving scalable data collection without physical hardware
- **Benchmarking imitation learning (ACT/Diffusion Policy) policies** trained on simulation-generated data against manually collected real-world demonstrations, quantifying the sim-to-real performance gap

Robotics Embedded Systems Lab 02/2026 – Present
University of Southern California

- **Distributed Multi-Robot Coordination:** Researching **Distributed Sequential Greedy Assignment (DSGA)** frameworks for online mapping and exploration, focusing on achieving computational **optimization in large-scale robot fleets**
- **Constrained Informative Planning:** Investigating **matroid-constrained submodular maximization** to coordinate autonomous agents in cluttered environments, integrating and ensuring "**sim-to-real**" consistency

Autonomous Networks Research Group 10/2025 – Present
University of Southern California under Prof. Bhaskar Krishnamachari

- **Conceptualizing and benchmarking** novel task-scheduling algorithms to **minimize execution latency** in edge networks
- **Synthesizing a scalable, Python-based simulation framework** from the ground up to model and validate the performance of complex, DAG-based computational workflows across thousands of distributed nodes
- **Implementing rigorous performance profiling** for mobility aware scenarios, quantifying tradeoffs in latency vs. throughput and validating system stability under variable network bandwidths, compute rates, and uplink/downlink delay

High Energy and Dendrite Free Micro Batteries 08/2021 – 05/2022
Braun Group, Materials Science and Engineering, UIUC, United States

- **Pioneered** a high-energy micro-battery (300μm) using a silicon foam anode to overcome the volumetric expansion and low-density limitations of conventional materials, **achieving a 30% higher energy density** than the graphite-anode baseline
- **Demonstrated** breakthrough cell stability and suppressed dendrite formation by **maintaining 95% of initial capacity after charge/discharge cycles**, a performance metric validated through rigorous potentiostatic analysis

Development of Solid State Ion Batteries 06/2021 – 07/2021
Nano Composite Research Lab, Korea University, Republic of Korea

- **Characterized novel Li/Na-based solid-state batteries** to address the safety limitations of traditional batteries.
- Diagnosed critical performance bottlenecks by employing **Electrochemical Impedance Spectroscopy (EIS) to quantify interfacial resistance**, a key obstacle limiting ionic conductivity in solid-state systems.
- **Conducted microstructural failure analysis using Scanning Electron Microscopy (SEM)** to identify physical defects at the electrode-electrolyte interface, directly informing improvements to the cell fabrication process.
- **Accelerated research lifecycle** by automating key processes: programmed a spin coater for precise, repeatable deposition and developed a script to parse and filter charge/discharge data points, eliminating manual data handling.

WORK EXPERIENCE

Ericsson 08/2023 – 06/2025

Software Engineer / Innovation Team Driver

- **Accelerated pre-production bug escapes by 25%** by architecting and deploying an intelligent, Kubernetes-native CI/CD pipeline, automating entire test and validation framework for the 5G Packet Core
- **Boosted end-to-end system test coverage by 40%** and **cut manual QA cycle** by developing 150+ functional and integration tests in Python and Erlang that auto-triaged failures, contributing to CI/CD
- **Optimized core 5G mobility functions (AMF/SMF) by analyzing network traffic patterns** to identify and eliminate critical performance bottlenecks, resulting in a reduction in packet latency and a 6% increase in user session throughput

Samsung Electronics 07/2022 – 08/2022

Global Intern

- **Drove a projected increase in FEOL process yield** by **diagnosing** a critical, previously overlooked failure pattern in the redistribution layer through a rigorous root cause analysis of wafer defect data
- **Expedited partner integration** by translating complex semiconductor packaging protocols into a new repository of concise technical briefs, eliminating a key communication bottleneck between engineering and partner relations

PROJECTS AND CONTRIBUTIONS

DAG Profiler 10/2025 – 03/2026

Lead Developer | https://pypi.org/project/dagprofiler/

- **Published an open-source Python library** for profiling DAGs, to identify execution bottlenecks in complex workflows
- **Developed automated performance metrics collection** to track node execution time, memory usage, and dependency
- **Illustrated a modular API** to ensure compatibility with popular orchestration frameworks, streamlining integration

Log Analyzer 08/2025 – 11/2025

Personal Project

- **Architected scalable, end-to-end observability platform** to ingest, process, and visualize high-volume log streams
- **Engineered a fault-tolerant ingestion pipeline** using a high-throughput Go endpoint and a distributed Python/Celery queue (RabbitMQ), demonstrating resilience against worker failure and processing backlogs
- **Delivered a real-time analytics dashboard** (Next.js) with JWT-based auth that enabled immediate anomaly detection, triggering alerts to Slack and cutting mean-time-to-resolution (MTTR) for system errors

PROFESSIONAL SKILLS

Microstructure Analysis
OOF2, LAMMPS, Quantum Espresso, SEM, OM, EIS, Raman Spectroscopy, Potentiostat, DSC, LVDT

Languages/Tools
Bash, C++, Dart, Erlang, Flutter, HTML/CSS, Java, JavaScript, Python, PyTorch, React, REST API, SQL (MySQL, PostgreSQL)

Cloud/Infrastructure
AWS, CI/CD, Docker, Firebase, Git/GitHub, Kubernetes

Microfabrication and Microstructure Analysis
Oxidation, Photolithography, Etching, Cleanroom Experience, Glovebox Experience, OOF2, LAMMPS, Quantum Espresso, SEM, OM, EIS, Raman Spectroscopy, Potentiostat, DSC, LVDT

AWARDS

Dean's List

University of Illinois at Urbana-Champaign

- Award granted to top 20 percentile students in the university
- Received on 2020 fall and 2021 spring semesters

PUBLICATION

Yoonjae H. (2026). Profiling and Scheduling Complex O-RAN Applications Across the 5G Edge and Cloud. ICCCN
Accepted for Publication